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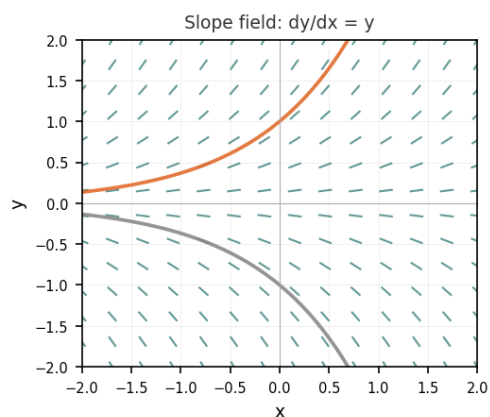
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## Learning Objectives

- Read and interpret slope fields; sketch solution curves through given points
- Solve separable differential equations with initial conditions
- Model exponential growth and decay by setting up and solving  $dP/dt = kP$
- Apply Newton's Law of Cooling and the logistic growth model

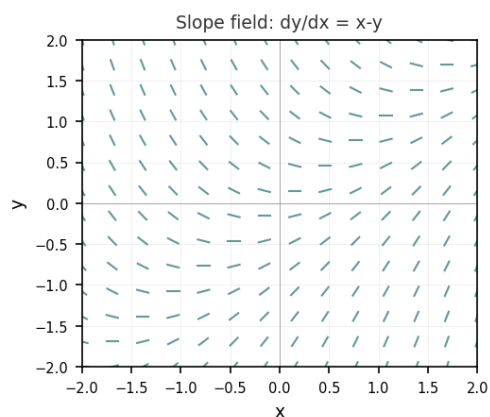
For separable DEs: separate variables completely before integrating. Always apply the initial condition AFTER finding the general solution.

**1. The slope field below is for  $dy/dx = y$ . Sketch the solution curve through  $(0, 1)$ . What is the general solution?**



Answer: \_\_\_\_\_

**2. Analyze the slope field below ( $dy/dx = x - y$ ). Describe where slopes are zero, positive, and negative.**



Answer: \_\_\_\_\_



3. Solve the separable differential equation with initial condition.

$$\frac{dy}{dx} = 3x^2y, \quad y(0) = 2$$

Answer: \_\_\_\_\_

4. Solve the separable differential equation.

$$\frac{dy}{dx} = \frac{x^2}{y^2}, \quad y(0) = 1$$

Answer: \_\_\_\_\_

5. Solve the separable DE.

$$\frac{dy}{dx} = e^{x-y} = \frac{e^x}{e^y}$$

Answer: \_\_\_\_\_

6. A bacterial population grows at a rate proportional to its size. Write and solve the differential equation. The population doubles in 3 hours, starting at  $P_0 = 500$ .

$$\frac{dP}{dt} = kP, \quad P(0) = 500, \quad P(3) = 1000$$

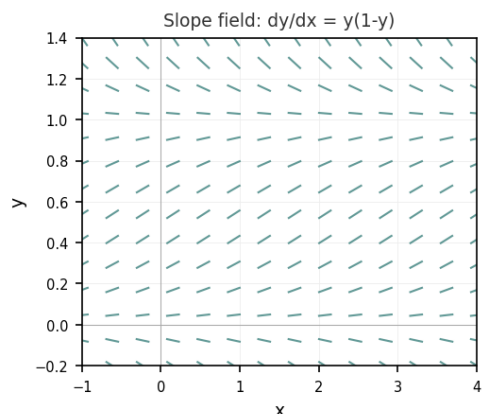
Answer: \_\_\_\_\_

7. Newton's Law of Cooling:  $dT/dt = -k(T - T_{\text{room}})$ . A cup of coffee at  $90^\circ\text{C}$  cools in a  $20^\circ\text{C}$  room. After 5 minutes, it is  $70^\circ\text{C}$ . Find  $T(t)$  and when it reaches  $40^\circ\text{C}$ .

$$\frac{dT}{dt} = -k(T - 20), \quad T(0) = 90, \quad T(5) = 70$$

Answer: \_\_\_\_\_

8. The logistic growth model is  $dP/dt = rP(1 - P/K)$ . With  $r = 1$  and  $K = 1$ , the slope field is shown. Identify equilibrium solutions and describe population behavior.



Answer: \_\_\_\_\_



9. Solve the first-order linear differential equation.

$$\frac{dy}{dx} + 2y = 4x, \quad y(0) = 1$$

Answer: \_\_\_\_\_

10. Verify that  $y = x^2 + Ce^{-x}$  is a solution to  $dy/dx + y = 2x + x^2$ .

$$\frac{dy}{dx} + y = 2x + x^2$$

Answer: \_\_\_\_\_

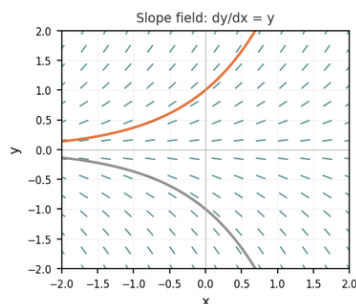




Problems 1–2 are slope field interpretation — build visual intuition before algebra. Problem 6 (bacterial growth): emphasize writing the DE first, then solving. Problem 8 (logistic): equilibrium analysis is more important than the explicit solution.

## Solutions

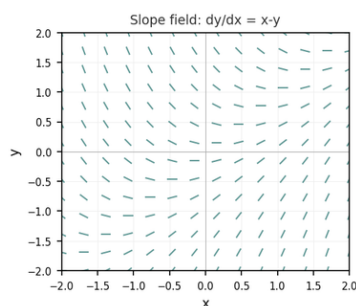
1. The slope field below is for  $dy/dx = y$ . Sketch the solution curve through  $(0, 1)$ . What is the general solution?



- $dy/dx = y$  is a separable equation:  $dy/y = dx$ .
- Integrate both sides:  $\ln|y| = x + C \rightarrow y = Ce^x$ .
- Initial condition  $y(0)=1$ :  $1=Ce^0=C \rightarrow C=1$ .
- Solution:  $y = e^x$ . The orange curve in the slope field.

**Answer:**  $y = Ce^x$ , through  $(0, 1)$ :  $y = e^x$

2. Analyze the slope field below ( $dy/dx = x - y$ ). Describe where slopes are zero, positive, and negative.



- Slopes are zero ( $dy/dx=0$ ) when  $x=y$  — along the line  $y=x$ .
- Slopes are positive when  $x > y$  (above the line  $y=x$ :  $dy/dx > 0$ ).
- Slopes are negative when  $x < y$ .
- This is a first-order linear DE. Integrating factor:  $e^x$ .
- General solution:  $y = x - 1 + Ce^{-x}$ .

**Answer:** Nullcline:  $x = y$  (zero slope along  $y = x$ )



3. Solve the separable differential equation with initial condition.

$$\frac{dy}{dx} = 3x^2y, \quad y(0) = 2$$

→ Separate:  $dy/y = 3x^2 dx$ .

→ Integrate:  $\ln|y| = x^3 + C$ .

→  $y = Ae^{x^3}$ . Apply  $y(0)=2$ :  $A=2$ .

→ Solution:  $y = 2e^{x^3}$ .

**Answer:**  $y = 2e^{x^3}$

4. Solve the separable differential equation.

$$\frac{dy}{dx} = \frac{x^2}{y^2}, \quad y(0) = 1$$

→ Separate:  $y^2 dy = x^2 dx$ .

→ Integrate:  $y^3/3 = x^3/3 + C$ .

→  $y^3 = x^3 + K$ . Apply  $y(0)=1$ :  $1=K$ .

→  $y^3 = x^3 + 1 \rightarrow y = (x^3 + 1)^{1/3}$ .

**Answer:**  $y = (x^3 + 1)^{1/3}$

5. Solve the separable DE.

$$\frac{dy}{dx} = e^{x-y} = \frac{e^x}{e^y}$$

→ Separate:  $e^y dy = e^x dx$ .

→ Integrate:  $e^y = e^x + C$ .

→  $y = \ln(e^x + C)$ .

**Answer:**  $e^y = e^x + C \Rightarrow y = \ln(e^x + C)$

6. A bacterial population grows at a rate proportional to its size. Write and solve the differential equation. The population doubles in 3 hours, starting at  $P_0 = 500$ .

$$\frac{dP}{dt} = kP, \quad P(0) = 500, \quad P(3) = 1000$$

→ Separable:  $dP/P = k dt \rightarrow \ln P = kt + C \rightarrow P = P_0 \cdot e^{kt}$ .

→  $P(0)=500$ :  $P_0=500$ .

→  $P(3)=1000$ :  $500e^{3k}=1000 \rightarrow e^{3k}=2 \rightarrow k=\ln 2/3$ .

→  $P(t) = 500 \cdot e^{(\ln 2/3)t} = 500 \cdot 2^{t/3}$ .

→ Find  $P(9)$ :  $500 \cdot 2^3 = 4000$  bacteria.

**Answer:**  $P(t) = 500 e^{(\ln 2/3)t}$

7. Newton's Law of Cooling:  $dT/dt = -k(T - T_{\text{room}})$ . A cup of coffee at  $90^\circ\text{C}$  cools in a  $20^\circ\text{C}$  room. After 5 minutes, it is  $70^\circ\text{C}$ . Find  $T(t)$  and when it reaches  $40^\circ\text{C}$ .

$$\frac{dT}{dt} = -k(T - 20), \quad T(0) = 90, \quad T(5) = 70$$

→ Let  $u = T - 20$ . Then  $du/dt = -ku \rightarrow u = u_0 e^{-kt}$ .

→  $T(t) = 20 + 70e^{-kt}$ . At  $t=5$ :  $70 = 20 + 70e^{-5k} \dots$  wait.

→  $T(5)=70 \rightarrow 70=20+70e^{-5k} \rightarrow 50=70e^{-5k} \rightarrow e^{-5k}=5/7$ .

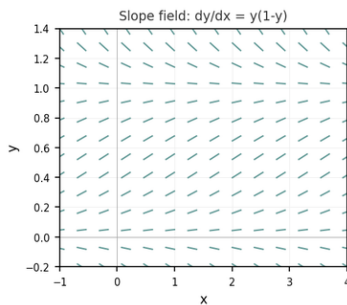
→  $k = -(1/5)\ln(5/7) = (1/5)\ln(7/5)$ .

→  $T=40$ :  $40=20+70e^{-kt} \rightarrow e^{-kt}=2/7 \rightarrow t=\ln(7/2)/k$ .

**Answer:**  $T(t) = 20 + 70e^{-kt}, \quad k = \frac{1}{5} \ln \frac{7}{5}$



8. The logistic growth model is  $dP/dt = rP(1 - P/K)$ . With  $r = 1$  and  $K = 1$ , the slope field is shown. Identify equilibrium solutions and describe population behavior.



- Set  $dP/dt=0$ :  $P(1-P)=0 \rightarrow P=0$  or  $P=1$ .
- $P=0$  is an unstable equilibrium (populations grow away from 0).
- $P=1$  (carrying capacity  $K=1$ ) is a stable equilibrium.
- Populations below  $K$  increase toward  $K$ ; populations above  $K$  decrease toward  $K$ .
- S-shaped (sigmoidal) solution curves are visible in the slope field.

**Answer:**  $P = 0$  (unstable),  $P = 1$  (stable carrying capacity)

9. Solve the first-order linear differential equation.

$$\frac{dy}{dx} + 2y = 4x, \quad y(0) = 1$$

- Standard form:  $dy/dx + P(x)y = Q(x)$ . Here  $P=2$ ,  $Q=4x$ .
- Integrating factor:  $\mu = e^{\int 2dx} = e^{2x}$ .
- Multiply:  $d/dx[e^{2x}y] = 4xe^{2x}$ .
- Integrate RHS by parts:  $\int 4xe^{2x}dx = 2xe^{2x} - e^{2x} + C$ .
- $e^{2x}y = 2xe^{2x} - e^{2x} + C \rightarrow y = 2x - 1 + Ce^{-2x}$ .
- $y(0)=1$ :  $1=0-1+C \rightarrow C=2$ .  $y=2x-1+2e^{-2x}$ .

**Answer:**  $y = 2x - 1 + 2e^{-2x}$

10. Verify that  $y = x^2 + Ce^{-x}$  is a solution to  $dy/dx + y = 2x + x^2$ .

$$\frac{dy}{dx} + y = 2x + x^2$$

- Compute  $dy/dx$ :  $dy/dx = 2x - Ce^{-x}$ .
- LHS =  $dy/dx + y = (2x - Ce^{-x}) + (x^2 + Ce^{-x})$ .
- $= 2x - Ce^{-x} + x^2 + Ce^{-x} = 2x + x^2$ .
- LHS = RHS =  $2x + x^2$ . ✓  $y = x^2 + Ce^{-x}$  is the general solution.

**Answer:** Verified: LHS =  $2x + x^2$  = RHS

